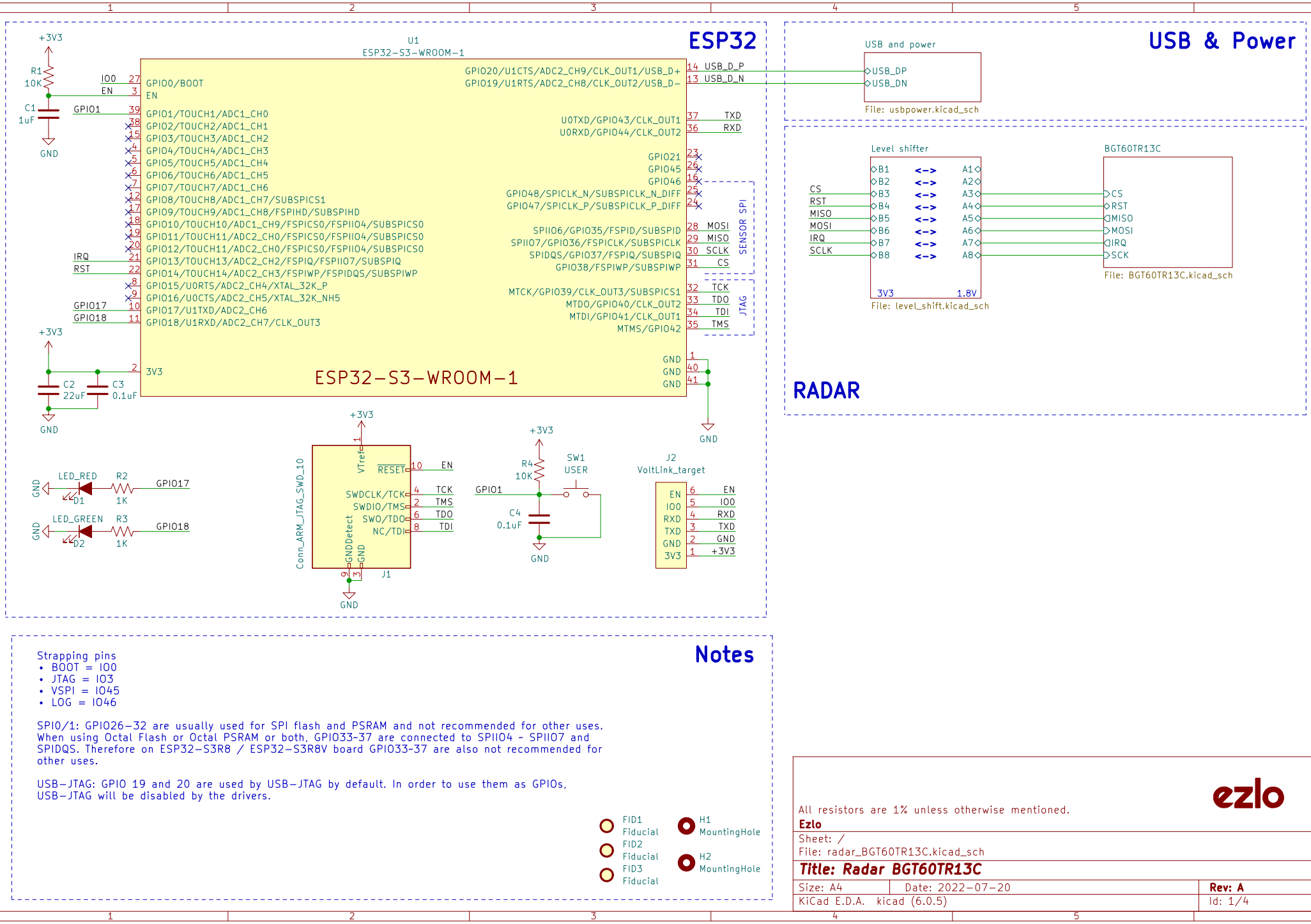
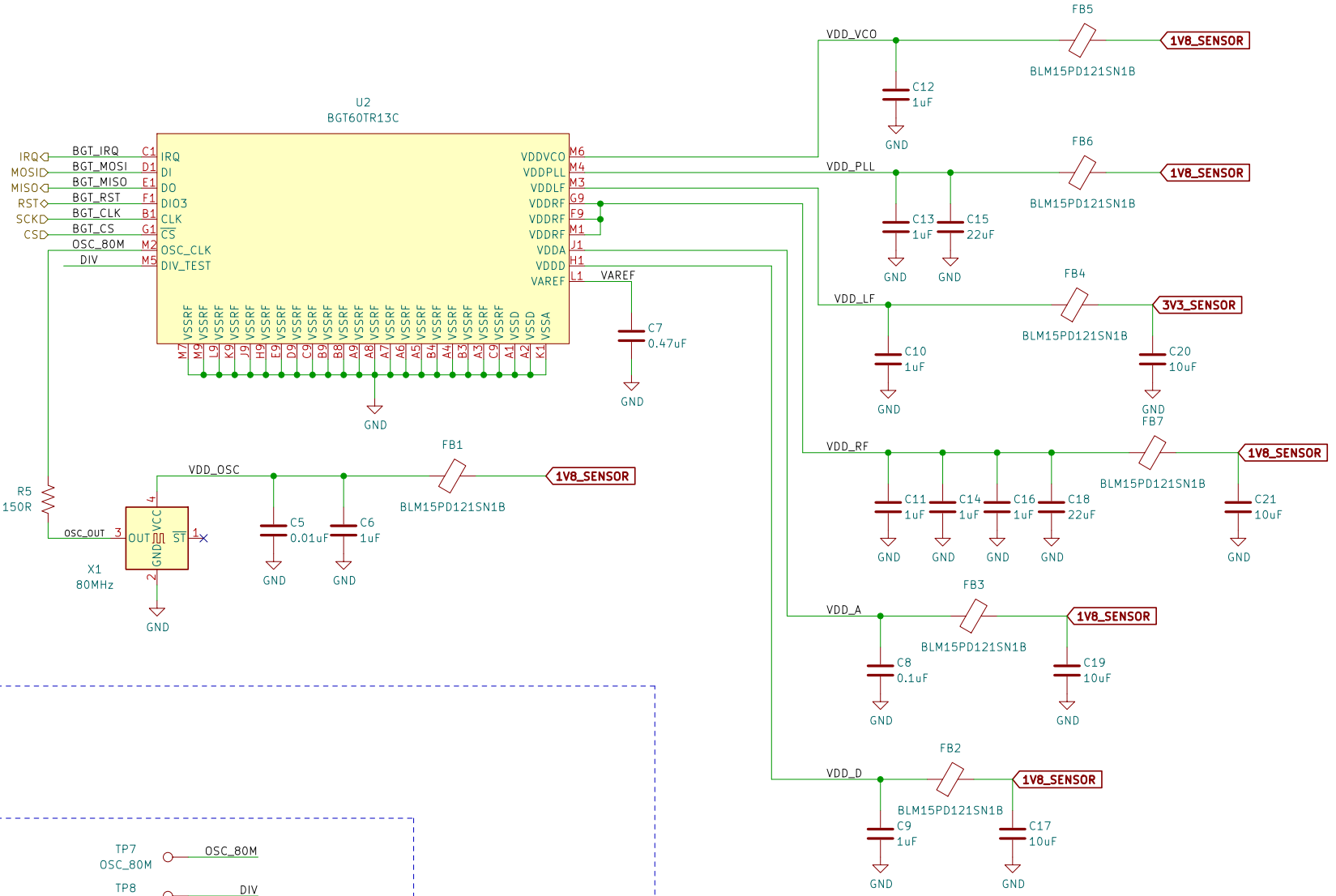
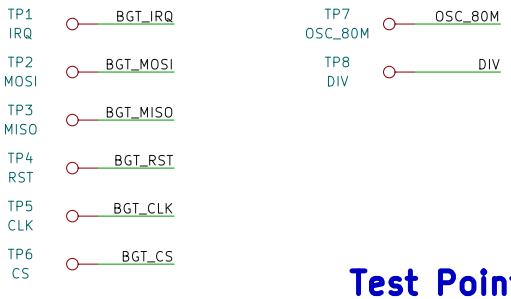

ESP32
U1
ESP32-S3-WROOM-1
ESP32-S3-WROOM-1
+3V3
R1
10K
C1
1uF
GND
IO0
EN
GPIO1
GPIO0/BOOT
EN
GPIO1/TOUCH1/ADC1_CH0
GPIO2/TOUCH2/ADC1_CH1
GPIO3/TOUCH3/ADC1_CH2
GPIO4/TOUCH4/ADC1_CH3
GPIO5/TOUCH5/ADC1_CH4
GPIO6/TOUCH6/ADC1_CH5
GPIO7/TOUCH7/ADC1_CH6
GPIO8/TOUCH8/ADC1_CH7/SUBSPICS1
GPIO9/TOUCH9/ADC1_CH8/FSPIHD/SUBSPIHD
GPIO10/TOUCH10/ADC1_CH9/FSPICS0/FSPII04/SUBSPICS0
GPIO11/TOUCH11/ADC2_CH0/FSPICS0/FSPII04/SUBSPICS0
GPIO12/TOUCH11/ADC2_CH0/FSPICS0/FSPII04/SUBSPICS0
GPIO13/TOUCH13/ADC2_CH2/FSPIQ/FSPII07/SUBSPIQ
GPIO14/TOUCH14/ADC2_CH3/FSPIWP/FSPIDQS/SUBSPIWP
GPIO15/U0RTS/ADC2_CH4/XTAL_32K_P
GPIO16/U0CTS/ADC2_CH5/XTAL_32K_NH5
GPIO17/U1TXD/ADC2_CH6
GPIO18/U1RXD/ADC2_CH7/CLK_OUT3
IRQ
RST
GPIO17
GPIO18
3V3
GPIO20/U1CTS/ADC2_CH9/CLK_OUT1/USB_D+
GPIO19/U1RTS/ADC2_CH8/CLK_OUT2/USB_D-
U0TXD/GPIO43/CLK_OUT1
U0RXD/GPIO44/CLK_OUT2
GPIO21
GPIO45
GPIO46
GPIO48/SPICLK_N/SUBSPICLK_N_DIFF
GPIO47/SPICLK_P/SUBSPICLK_P_DIFF
SPII06/GPIO35/FSPID/SUBSPID
SPII07/GPIO36/FSPICLK/SUBSPICLK
SPIDQS/GPIO37/FSPIQ/SUBSPIQ
GPIO38/FSPIWP/SUBSPIWP
MTCK/GPIO39/CLK_OUT3/SUBSPICS1
MTDO/GPIO40/CLK_OUT2
MTDI/GPIO41/CLK_OUT1
MTMS/GPIO42
GND
GND
GND
USB_D_P
USB_D_N
TXD
RXD
MOSI
MISO
SCLK
CS
TCK
TDO
TDI
TMS
SENSOR SPI
JTAG
C2
22uF
C3
0.1uF
LED_RED
D1
R2
1K
LED_GREEN
D2
R3
1K
Conn_ARM_JTAG_SWD_10
VTref
RESET
SWDCLK/TCK
SWDIO/TMS
SWO/TDO
NC/TDI
GNDDetect
J1
R4
10K
SW1
USER
C4
0.1uF
J2
VoltLink_target
EN
IO0
RXD
TXD
GND
3V3
USB & Power
USB and power
USB_DP
USB_DN
File: usbpower.kicad_sch
RADAR
Level shifter
B1
B2
B3
B4
B5
B6
B7
B8
A1
A2
A3
A4
A5
A6
A7
A8
3V3
1.8V
File: level_shift.kicad_sch
BGT60TR13C
CS
RST
MISO
MOSI
IRQ
SCK
File: BGT60TR13C.kicad_sch
Notes
Strapping pins
• BOOT = IO0
• JTAG = IO3
• VSPI = IO45
• LOG = IO46
SPI0/1: GPIO26-32 are usually used for SPI flash and PSRAM and not recommended for other uses. When using Octal Flash or Octal PSRAM or both, GPIO33~37 are connected to SPII04 ~ SPII07 and SPIDQS. Therefore on ESP32-S3R8 / ESP32-S3R8V board GPIO33~37 are also not recommended for other uses.
USB-JTAG: GPIO 19 and 20 are used by USB-JTAG by default. In order to use them as GPIOs, USB-JTAG will be disabled by the drivers.
FID1
Fiducial
FID2
Fiducial
FID3
Fiducial
H1
MountingHole
H2
MountingHole
All resistors are 1% unless otherwise mentioned.
Ezlo
Sheet: /
File: radar_BGT60TR13C.kicad_sch
Title: Radar BGT60TR13C
Size: A4
Date: 2022-07-20
Rev: A
KiCad E.D.A. kicad (6.0.5)
Id: 1/4

RADAR SENSOR





Test Points

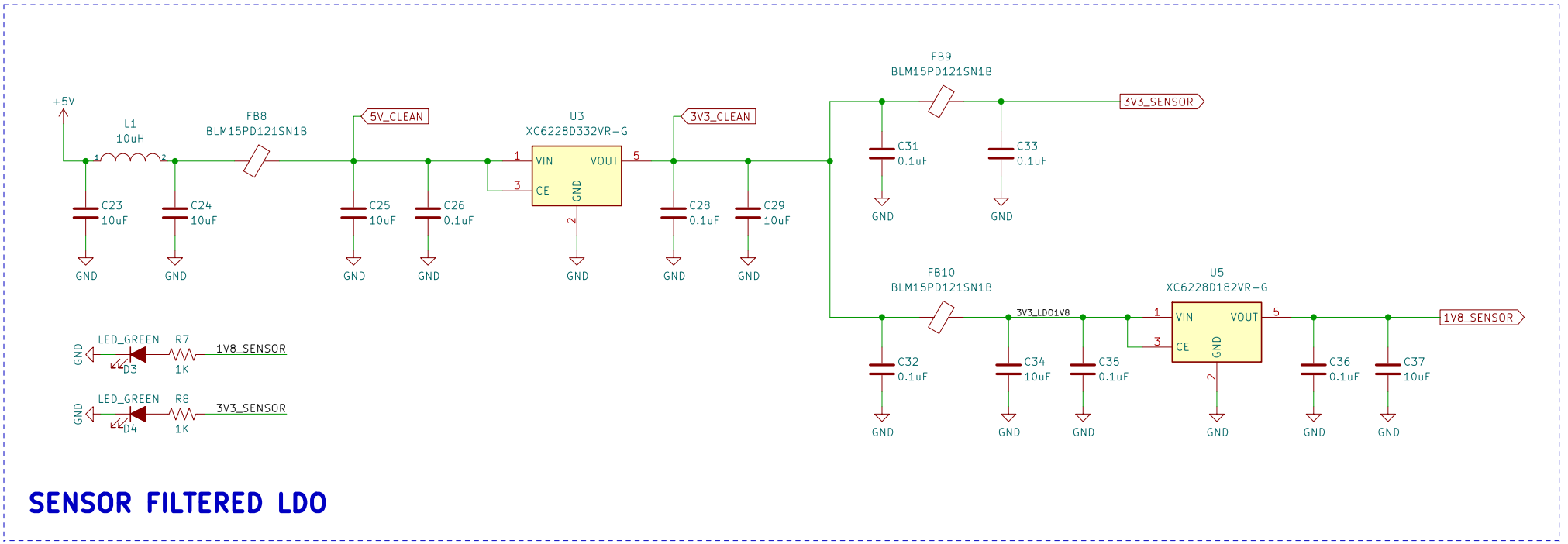
All resistors are 1% unless otherwise mentioned.

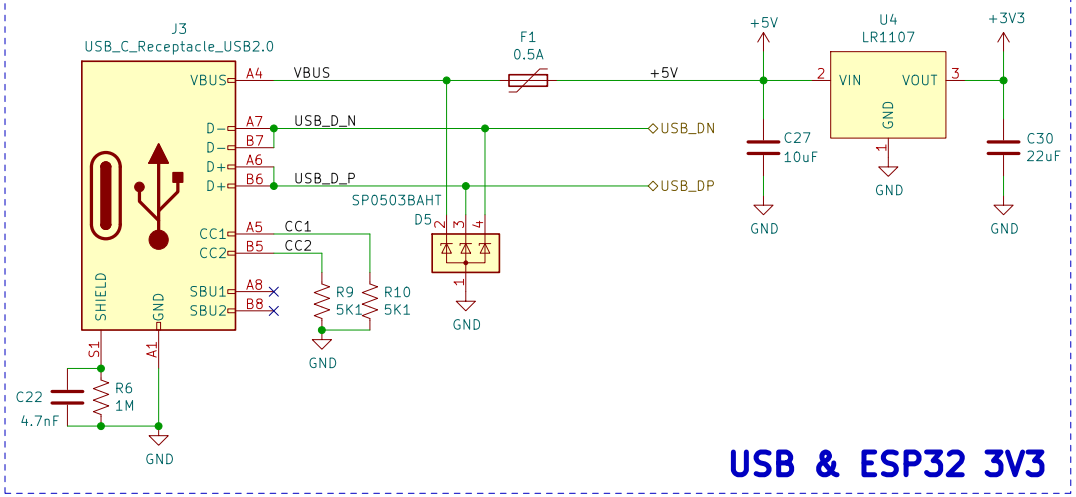
Ezlo

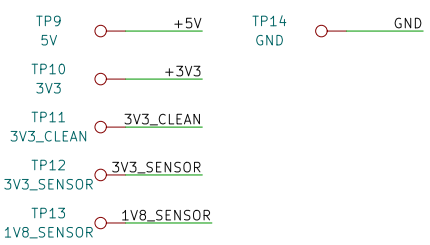
Sheet: /BGT60TR13C/
File: BGT60TR13C.kicad_sch

Title:

Size: A4	Date:	**Rev: A**
KiCad E.D.A. kicad (6.0.5)		Id: 3/4







All resistors are 1% unless otherwise mentioned.

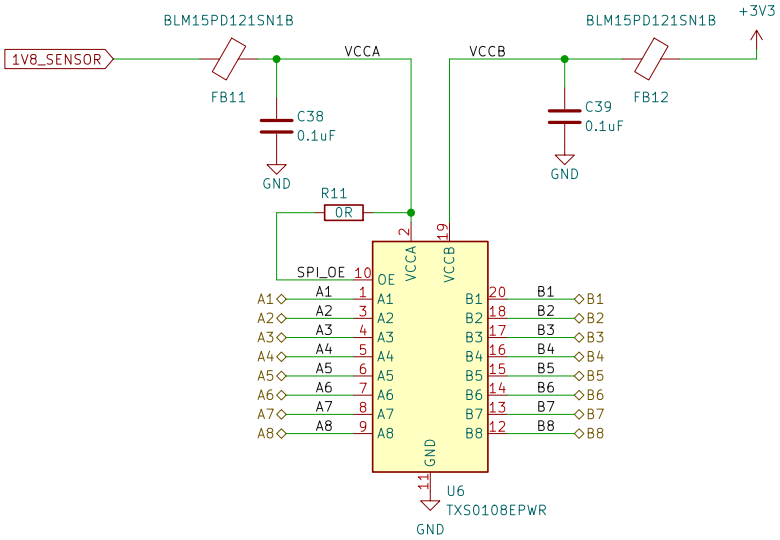
Ezlo

Sheet: /USB and power/
File: usbpower.kicad_sch

Title:

Size: A4	Date:	**Rev: A**
KiCad E.D.A. kicad (6.0.5)		Id: 4/4

Level Shifter



1.4 V to 3.6 V on A port and 1.65 V to 5.5 V on B port (VCCA ≤ VCCB)!

All resistors are 1% unless otherwise mentioned.

Ezlo

Sheet: /Level shifter/
File: level_shift.kicad_sch

Title:

Size: A4	Date:	**Rev: A**
KiCad E.D.A. kicad (6.0.5)		Id: 5/4